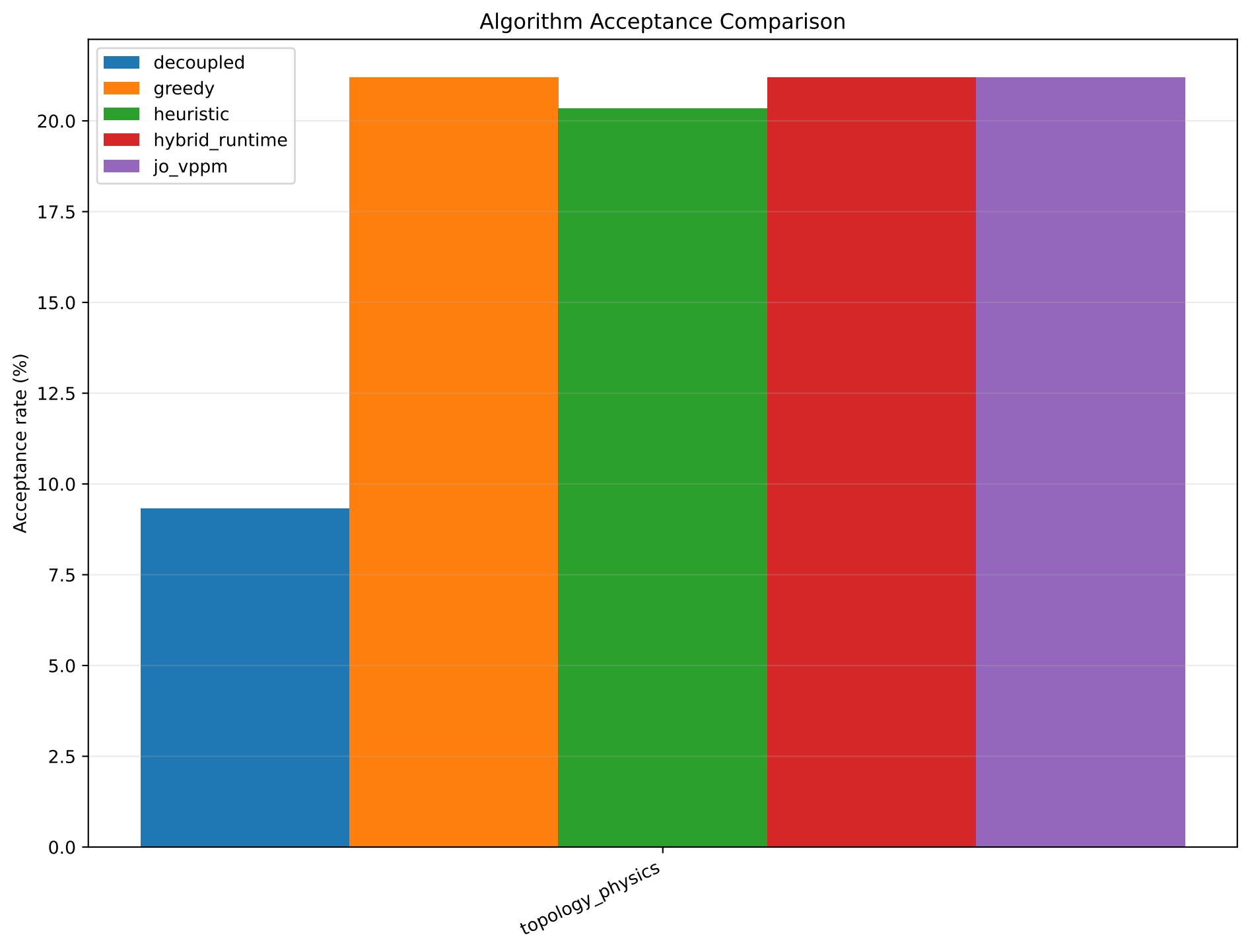
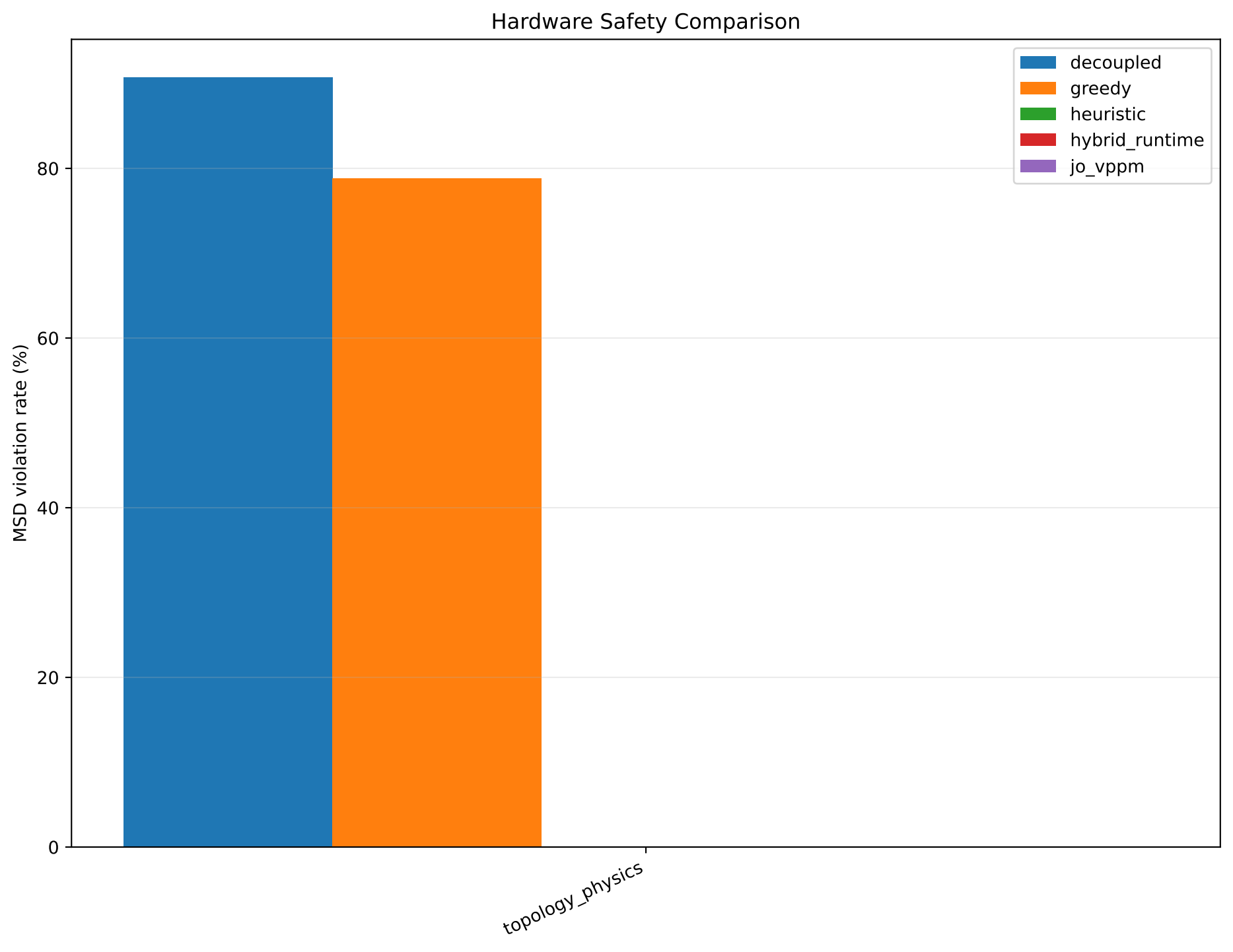
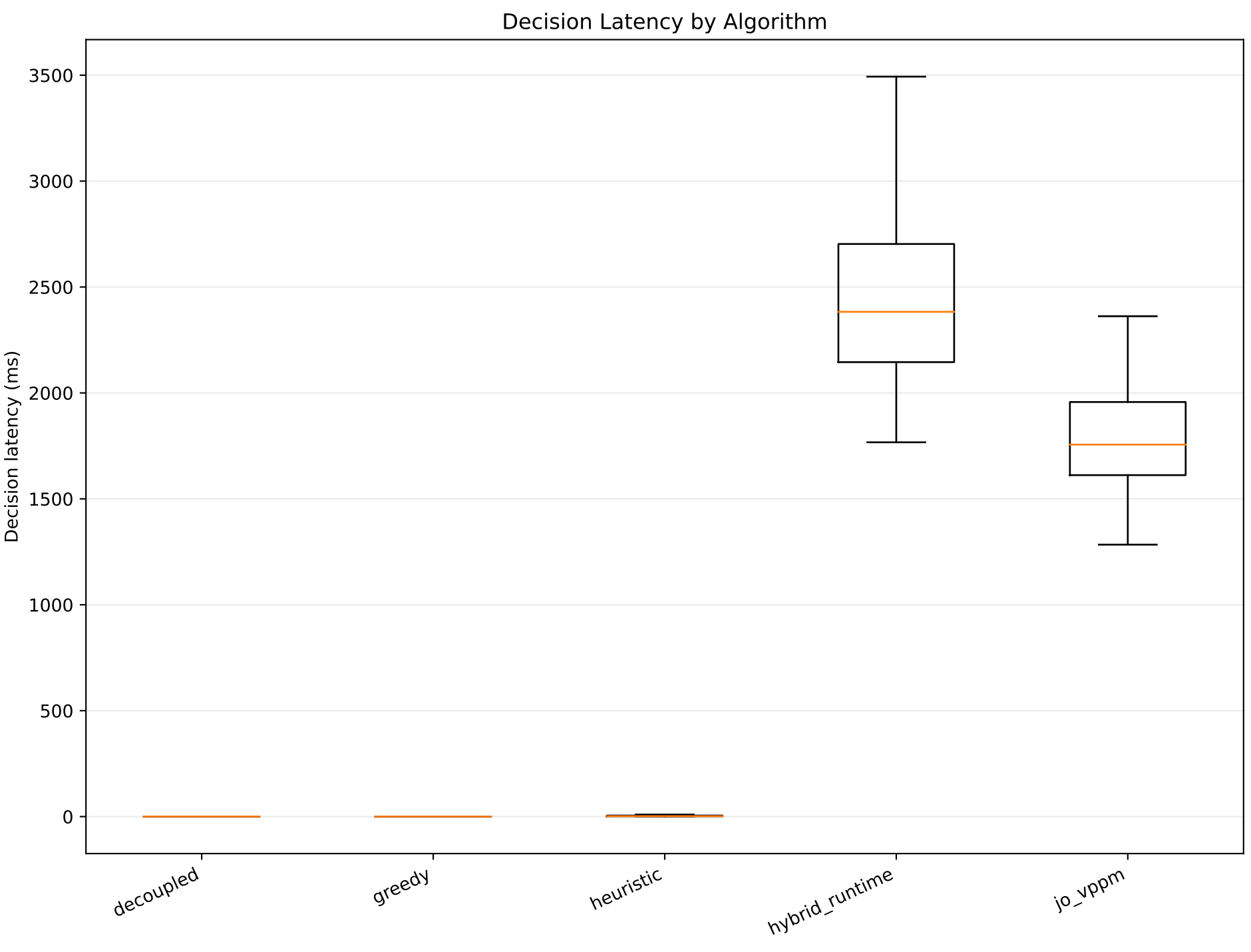
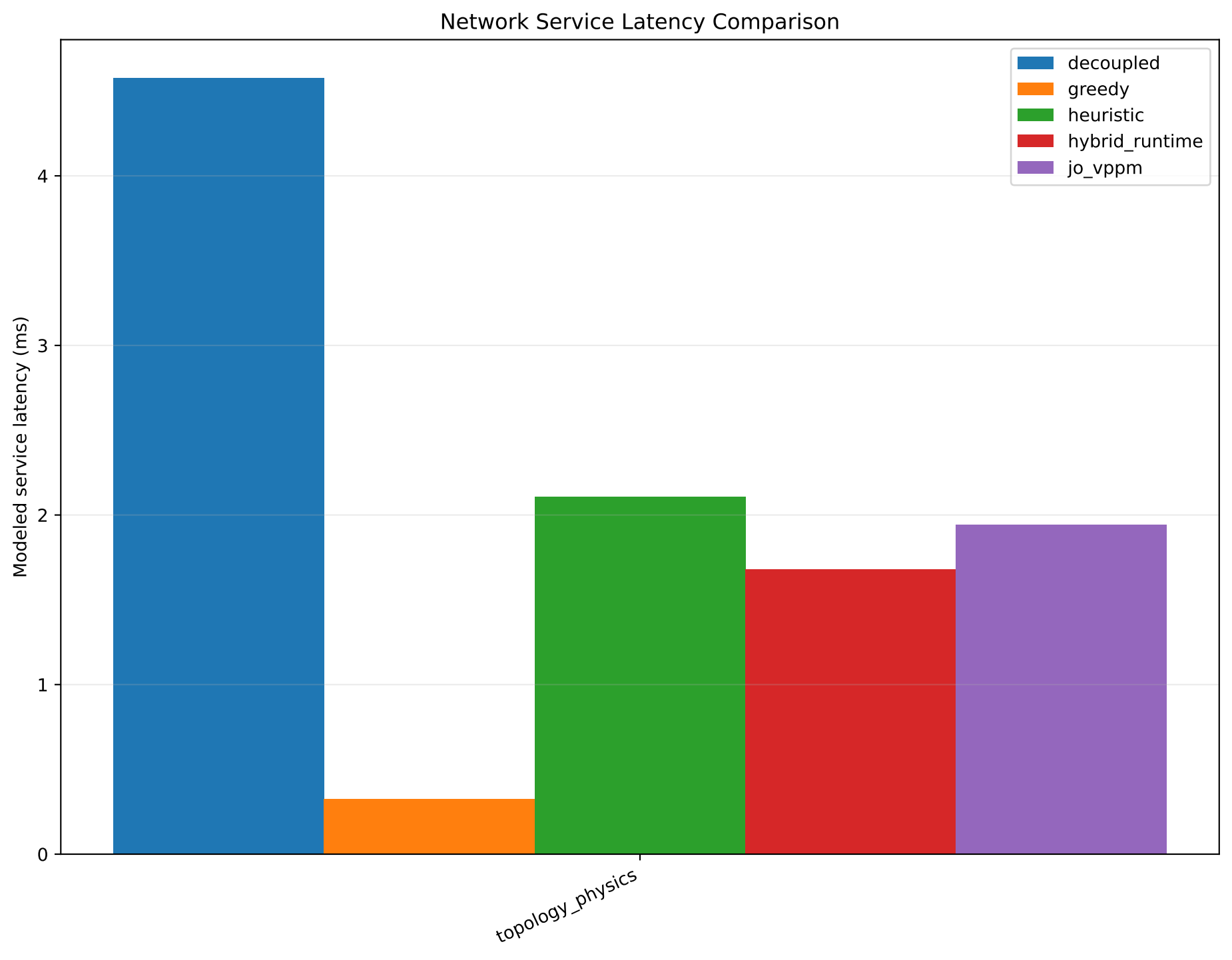
3S-COM Testbed Scenario Report - Scenario 3 - Topology and Physics Stress

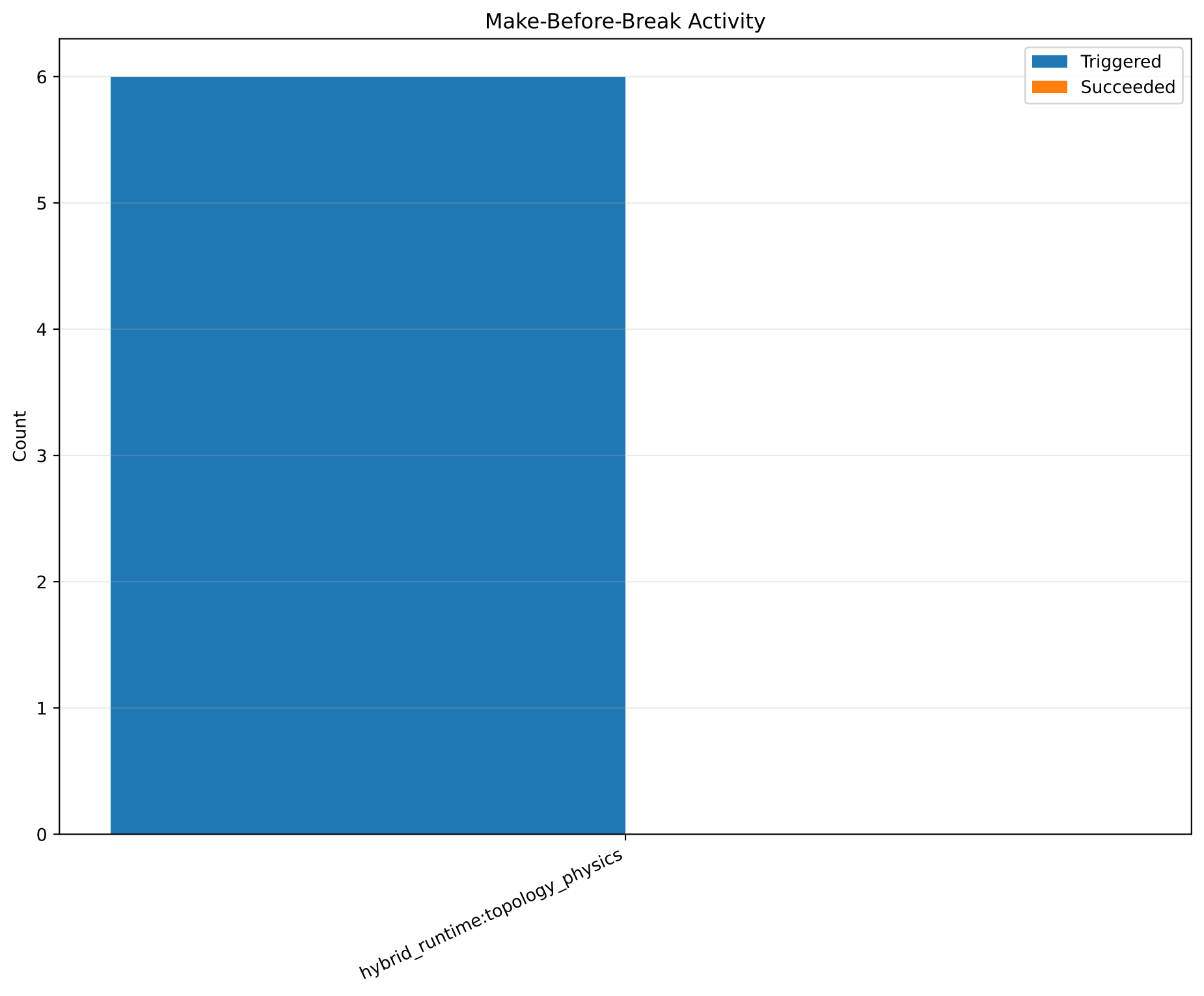
Scenario	Algorithm	Requests	Accept	No safe	MSD viol	Decision ms	Service ms	Timeouts	Migrations	MSD drops
o 3 - Topology and P	hybrid_runtime	118	21.2%	78.8%	0.0%	3992.8	1.68	0	6	0
o 3 - Topology and P	jo_vppm	118	21.2%	78.8%	0.0%	1868.6	1.94	0	0	0
o 3 - Topology and P	heuristic	118	20.3%	79.7%	0.0%	9.7	2.11	0	0	0
o 3 - Topology and P	greedy	118	21.2%	0.0%	78.8%	0.4	0.32	0	0	0
o 3 - Topology and P	decoupled	118	9.3%	0.0%	90.7%	0.7	4.57	0	0	0


Algorithm Acceptance Comparison
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
Acceptance rate (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
topology_physics


Hardware Safety Comparison
MSD violation rate (%)
0
20
40
60
80
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
topology_physics


Decision Latency by Algorithm
Decision latency (ms)
3500
3000
2500
2000
1500
1000
500
0
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Network Service Latency Comparison
Modeled service latency (ms)
0
1
2
3
4
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
topology_physics


Make-Before-Break Activity
Triggered
Succeeded
Count
0
1
2
3
4
5
6
hybrid_runtime:topology_physics